

DraftFM: Zero-Shot Drafting of Unseen *Magic: The Gathering* Sets from Public Card Features

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Abstract

Draft assistants for *Magic: The Gathering* need weeks of post-release human pick data before they can model a new set. We train DraftFM, a pick model that represents every card as a frozen vector of public information: structured Scryfall features plus a sentence embedding of its rules text. It holds no card- or set-specific parameters, so it can score a set the moment its card list goes public.

We first train a model on 149.4M human picks from 28 sets of 17Lands public data to pick the architecture and hyperparameters. On three sets it never trained on, that model agrees with expert human picks 54.3% of the time. To put that number in context: a model trained directly on one of those three sets, and so allowed to actually see it, marks the best any model can do there. DraftFM, trained on zero picks from the set it is scoring, still reaches 78.6% of what that directly-trained model achieves (the pre-registered normalized score, §5). It also beats every published number we could find that a real deployment could have used on release day. Each of those numbers comes from a different test set, so we treat them as reference points rather than a head-to-head ranking (Table 2).

A follow-up experiment on the same three sets tried a simpler architecture: one without the added component that reads the whole set’s card list. The simpler model scored no worse, within seed noise, so we deployed it instead. This deployed recipe, which we call F-full, has 1.6M parameters. It produces the paper’s headline result: a single pre-registered pass over the first public snapshot of MSH, a licensed-IP set untouched by any training or tuning in this pipeline, reaching 57.0% expert top-1 agreement. We release the code, the frozen evaluation protocol, model weights, and per-pick prediction files.

1 Introduction

Drafting in *Magic: The Gathering* is a sequential game of imperfect information: eight players pass packs of cards, each privately assembling a deck one pick at a time. Strong pick models exist. Supervised models trained on human draft logs reach 48.7–71% agreement with human picks within a set [5, 6, 10] (a general, not expert-filtered, pick population; experts are easier to predict, scoring several points higher), but they are set-specific. Each model encodes cards as vocabulary indices, learns from picks made in that set, and transfers nowhere. Because public draft logs appear on a delay, a data-driven assistant for a new set cannot exist until roughly two weeks after release [5]. On release day, the state of the art is a spreadsheet of hand-tuned expert ratings [7].

This paper asks how far pure transfer can go. DraftFM is a single pick model trained across every public 17Lands draft set. To DraftFM, a card is nothing but a frozen vector of public information: structured features parsed from its Scryfall record, plus a sentence embedding of its rules text. The model has no set-identity parameters and no per-card weights. Whatever it knows about a set it must infer from the cards in the pack and the drafter’s accumulating pool, not from a memorized identity. That inference transfers to a set released tomorrow. A memorized card vocabulary cannot.

A model like this invites an obvious failure mode: report a number that was quietly tuned on the very test set it claims never to have seen. We designed the evaluation against exactly that risk. The protocol (§5) was frozen in a tagged public commit before the test set existed. That test set is MSH, a licensed-IP set whose public draft data had not yet been published at the freeze. No training, tuning, or metric choice touches MSH pick data; the headline number comes from a single pre-registered inference pass over the first public snapshot. All model-selection decisions instead happen on three development sets, BRO, TMT, and SOS, held out of training. We chose that trio so our numbers are directly comparable to the only published multi-set zero-shot benchmark built on public features [3] (same BRO holdout) and to our own strongest per-set models.

On the development trio, DraftFM agrees with expert picks 54.3% of the time (48.9% / 57.5% / 56.5% on BRO / TMT / SOS). To read that number, it helps to know the best any model can do on one of these sets: train a model directly on the set, so it actually sees the set’s own picks. That directly-trained model marks the ceiling. DraftFM, despite training on zero picks from the target set, reaches 78.6% of that ceiling (the pre-registered normalized score, identical held-out picks, §5).

DraftFM also clears every published number we could find that a real deployment could have used on release day. Three reference points: an expert-tuned rating system at 44.5% [10] on M19; an hour-zero rarity heuristic at 42.8% (measured on SOS as a rehearsal stand-in, not per dev set); and zero-shot GPT-4o at 43% [2] on NEO. Each of these numbers comes from a different test set than DraftFM’s, so Table 2 presents them as reference points rather than a head-to-head ranking. The one comparison that does share a test set is the rarity heuristic against DraftFM’s own SOS number (56.5%). On MSH itself, the frozen evaluation gives 57.0% expert top-1 agreement.

Contributions:

- **A cross-set model and recipe.** A two-tower architecture over frozen card features. A pre-registered comparison tests a 2.2M-parameter design that also reads the full set’s card list against a simpler 1.6M-parameter version that drops that component; the simpler version wins and is what ships (we call it A-noctx, §7). It trains on 149.4M picks in about 1.8 hours on a single consumer workstation (§4).
- **A frozen zero-shot benchmark.** A pre-registered, tamper-evident protocol for day-one draft evaluation, plus the first multi-set zero-shot numbers that use no post-release information of any kind (§5, §6).
- **A scaling study.** Zero-shot accuracy as a function of training-set count from 1 to 28 sets, with set-composition probes and zero-shot seed bands at two rungs (§6; the bands bound the curve’s seed noise rather than fully characterize it, see caveat there).
- **A licensed-IP shift analysis.** Pre-registered experiments that isolate what text embeddings contribute when a set’s names and flavor come from a licensed universe rather than in-universe *Magic: The Gathering* (§7).
- **Release.** Code, the frozen protocol, data manifests, model weights, and per-pick prediction files for every number in this paper (Appendix B).



Figure 1: Emeritus of Ideation, an actual first pick from a real thirteen-card SOS pack (empty pool). The per-set ceiling model (Table 1) ranks it above every other card in the pack by a wide margin: real expected value¹ 8.61, seen in 65.1% of games it’s in, average last-seen-at pick 1.4, and >99% model dominance over the closest candidate in the pack (Quandrix Charm). SOS is a public development set, never the frozen MSH test set (§5); this figure illustrates the model’s output, not a headline result. Numbers regenerated by figures/make_verdict_panels.py.

2 Related work

Generalised card representations. Bertram, Fürnkranz, and Müller [3] established the problem of drafting a set the model never trained on. Their paper remains the only published quantitative result on it from multi-set, feature-based training (a separate LLM number exists, discussed below under “LLMs as drafters”). They replace one-hot card identities with concatenated feature vectors: hand-engineered attributes, a Sentence-BERT embedding [8] of card text (1306-d), a learned autoencoding of card images (1024-d), and 16 post-release usage statistics from 17Lands. This lives inside a contextual-preference-ranking model trained with a triplet loss. Two of their findings shape our design. Feature representations match one-hot accuracy within a set, so nothing is lost by generalising. And multi-set pretraining, not the representation alone, is what buys transfer to an unseen set. Pretrained on 13 sets and evaluated on held-out BRO, their best representation reaches 55.4% pick agreement. But that representation includes the usage-statistics block, which is computed from human play after a set’s release. The 55.4% number is therefore an anchor for generalising across *cards*, not for day-one deployment, where those statistics cannot yet exist. The same paper’s day-one-feasible representation, features only, reaches 33.6–35.6% on unseen sets under single-set training, and no multi-set, features-only number is published. Filling that gap, multi-set training on public card information only, is this paper’s headline experiment; we adopt their held-out BRO as a development set so the comparison uses the same holdout. A companion paper [4] improves the within-set triplet objective with an adapted InfoNCE loss but reports no unseen-set results.

Within-set pick models. Ward et al. [10] compare heuristic and neural drafters on M19. Two of their reference points matter most below the data-driven regime: an expert-tuned rating bot

¹*Expected value* is this per-set model’s own raw score for the card (higher is better; not a model input, and not the top-1 agreement metric reported elsewhere in this paper). *GIH%* and *ALSA* are 17Lands per-card statistics reported for context: *GIH%* is win rate in games where the card was drawn, and *ALSA* is the average pick number at which the card was last seen (lower means drafters value it more highly). *Model dominance* is this paper’s own pairwise measure, $\text{sigmoid}(\text{top EV} - \text{runner-up EV})$.

at 44.5% top-1 agreement, and a random agent at 22%. Their neural drafter (48.7%) and its successors define the within-set state of the art: the statistical-drafting MLPs deployed at statisticaldrafting.com [5] ($\sim 70\%$) and Czerner’s pudor transformer [6] (71%). The statistical-drafting architecture is also the direct lineage of the per-set ceiling models we compare against. All of these are one-hot models over a fixed card vocabulary, so none can score an unseen set. The deployment gap is explicit in prior work: statistical-drafting documents that no model can ship until 17Lands data appears, roughly two weeks post-release [5], and Draftsim’s day-one ratings are written by human experts [7].

LLMs as drafters. UrzaGPT [2] evaluates large language models on draft picks. Zero-shot GPT-4o reaches 43%, narrowly above the hour-zero rarity heuristic we measure (§6). We still mark it non-day-one in Table 2: NEO shipped years before GPT-4o’s training cutoff, so its pretraining corpus cannot be shown free of the set’s own post-release strategy discussion, a leak a frozen feature-only encoder cannot have. Providing full rules text in the prompt lowers GPT-4o’s accuracy further. Open 7–8B models fail to pick legal cards until LoRA-tuned on a million picks from the target set, after which they still trail small supervised MLPs at orders of magnitude more inference cost. We take the implied guidance: use a frozen text *embedding* as one feature block, not a language model as the policy.

Text-grounded transfer in card games. Cardsformer [11] encodes Hearthstone card descriptions with a language model so a gameplay policy generalises to unseen cards. It is the closest cross-game precedent for text-mediated transfer, applied to gameplay rather than drafting.

3 Data

Corpus. Training data is the complete 17Lands public draft-data corpus [1], spanning every set the model has to generalise across: 31 expansions from STX (April 2021) through SOS (June 2026), Premier (best-of-one) and Traditional (best-of-three) draft where published, 59 (set, format) files, roughly 160M pick rows. Each row is one human pick: the pack contents, the drafter’s pool, the card taken, and two coarse skill covariates (recent game-count bucket and win-rate bucket). Files were fetched once from the public S3 bucket, per the 17Lands usage guidelines (bulk downloads only), and frozen by sha256 and ETag in a data manifest (content hash 443bd25f72ee); the manifest is released with the paper.

Two public files are structurally excluded from the corpus, recorded in the corpus registry: OM1 (a pick-two format whose rows carry two picks and whose pool columns undercount them, and whose digital-only cards largely do not exist on Scryfall) and the Powered Cube (a curated 545-card singleton pool with no expansion structure). A third, the VOW QuickDraft file (human picks inside bot pods, off-distribution pack dynamics), is held out of the default corpus for a distributional reason rather than a structural one: it is one of the two pre-registered flexibility-clause extras (§5) that may join F-full’s training data if a dev-trio ablation shows a gain. Everything else is kept, including digital-only and remaster sets.

Normalization. The corpus spans three schema eras and several quirks, all normalised at ETL time: the 2021 files are gzipped tarballs rather than plain gzipped CSVs, and the oldest era reports match-based rather than game-based skill buckets (harmonised; identical in best-of-one). Five sets are missing the first pick of each draft in the source logs (an annotation, not an exclusion), and packs are 13, 14, or 15 picks depending on the set (a model input, §4). Card names are joined to

Scryfall by normalised name with an explicit alias table; unresolved names fail the build rather than featurizing as zeros. Per set we retain every valid pick and tag it with the drafter’s skill covariates: training filters no one out. Skill is a conditioning input (“decision + tag by venue”).

Splits. Within each (set, format) shard, drafts are split by a hash of the draft identifier (`crc32 % 1000 < 50`, $\sim 5\%$ validation). Early stopping uses only this within-training-set validation split. The *expert slice* used for all headline evaluation is picks by drafters with win-rate bucket ≥ 0.55 and game-count bucket ≥ 100 , the same filter the per-set ceiling models train and evaluate on, so ceiling comparisons are population-matched.

The held-out sets. Four sets never contribute a pick to model selection. BRO, TMT, and SOS are the development trio (§5): all model selection keys on zero-shot accuracy over these three, computed from *F-dev*, which trains on none of them. Once selection is frozen, the headline model (*F-full*) retraining on all 31 sets, dev trio included: only MSH stays unseen by every model in the battery. MSH, the test set, is stricter still: a registry-level gate (`EVAL_ONLY`) makes every training and feature-fitting code path refuse MSH pick data outright, and at the protocol freeze its public data did not yet exist. MSH *card* features from Scryfall are permitted. That is the zero-shot contract: public card information only, no human behavioral data of any kind. “Unseen” describes the set, not every card in it. BRO, TMT, and SOS packs include some ordinary reprints (mostly base-set staples and basic lands) that also appear in *F-dev*’s training corpus under other sets. This confers no advantage under the contract above: a reprint has no memorized identity to exploit, only the same frozen feature vector any novel card with identical features would get. But it does mean set-level and card-level novelty are not the same claim.

4 Method

4.1 Cards as frozen feature vectors

Every card is a fixed 775-dimensional vector computed from public information alone, frozen before training.

Structured features (391-d). Parsed from the card’s Scryfall record: mana value (scaled + bucketed one-hot, 10), mana-cost pips including hybrid/Phyrexian/X structure (10), colors (8), color identity (5), supertypes (3), card types (9), creature subtypes (top-128 vocabulary + an unmatched count, 129), power / toughness / loyalty each as (scaled, missing, star) triples (9), rarity (5), keyword abilities (166 slots + unmatched count, 167), layout including back-face flags for double-faced cards (10), and text-shape features: length, line count, and 24 regex flags over rules text for common functional categories (removal, card draw, counterspells, tokens, ramp, ...) (26). Double-faced and split cards take numeric blocks from the front face. Subtype and keyword vocabularies are counted over *training* sets only; a registry gate refuses to fit vocabularies on MSH, the frozen test set (§5). That guarantee covers MSH specifically. It does not mean the dev trio’s cards were excluded from every vocabulary in the release (§3), so calling the dev trio “zero-shot” is a claim about training picks, not about where the vocabulary came from. The resulting featurizer manifest is content-hashed (`e1313cadd03b`) and released; held-out cards featurize through the frozen manifest, with out-of-vocabulary subtypes and keywords absorbed by the unmatched counters. Scalars are clipped to $[0, 1]$; parse failures set a missing indicator, never a silent zero.

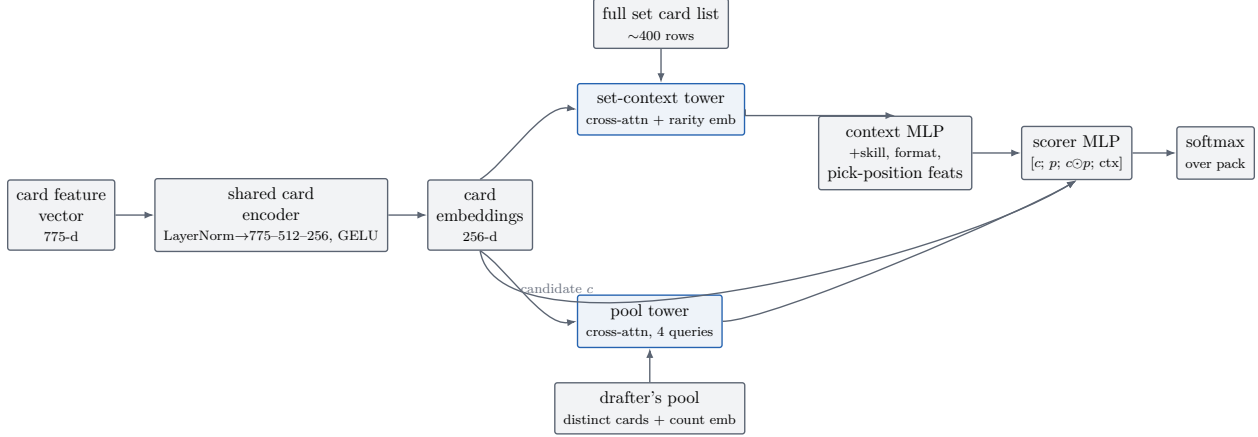


Figure 2: The two-tower scorer (§4.2). A shared card encoder embeds every card once per (set, format) step; a pool tower summarises the drafter’s held cards, a set-context tower summarises the full card list, and a per-candidate scorer combines the candidate embedding c , the pool summary p , their elementwise product, and the context vector into a softmax over the pack.

Text embedding (384-d). The card’s type line and rules text, embedded with a frozen `bge-small-en-v1.5` sentence encoder [12] (L_2 -normalised). The card’s own name (and, for legendary cards, its short name) is masked to a placeholder token before embedding, so the encoder cannot key on names it may know from pretraining, a deliberate handicap that matters for licensed-IP sets (§7). Parenthetical reminder text is kept: for a brand-new mechanic it is the only definition the model will ever see. Back faces are appended for double-faced cards.

4.2 Architecture

DraftFM is a two-tower scorer with 2.2M parameters (Figure 2). Its two design invariants operationalize the zero-shot contract (§3):

1. **No set identity.** There is no set embedding and no per-card parameter anywhere. The only way the model can know what set it is drafting is a *set-context tower* that reads the set’s full card list, plus the drafter’s accumulating pool. “Blue is the aggressive color in this set” must be inferred from the card list, not memorized as a set identifier. Whether the tower itself turns out to be necessary for that inference, or whether the invariant holds just as well without it, is addressed later in this section.
2. **Skill conditions the scorer, not the cards.** Card semantics are skill-invariant; the pick policy is not. Skill-bucket embeddings enter only the context pathway. Serving conditions on a fixed high-skill bucket; training sees every pick tagged with its drafter’s bucket.

A shared *card encoder* (LayerNorm $\rightarrow$ 775-512-256 MLP, GELU) maps feature vectors to $d=256$ embeddings. Because training batches are homogeneous in (set, format), the encoder runs once per step over the set’s full card list (~ 400 rows) and everything downstream gathers rows from that table. The *pool tower* represents the drafter’s pool as the set of distinct cards held, each embedding translated by a learned count embedding (counts capped at 8), pooled by cross-attention from four learned queries (4 heads); an empty pool attends over a learned null token. The *set-context tower* applies the same query-pooling to the entire card list (with a rarity embedding added) to produce

the set summary. A context MLP combines the set summary with skill and format embeddings, seven pick-position features (pack/pick indices, pool size), and four set-shape scalars (card-list size, picks-per-pack indicator) into a 64-d context. Each candidate c in the pack is scored by an MLP over $[c; p; c \odot p; \text{ctx}]$, where p is the pool summary and $c \odot p$ the elementwise interaction; a softmax over the pack’s real slots gives the pick distribution.

We test the set-context tower’s contribution directly by removing it and retraining (we call this variant A-noctx, Table 4, §7). It scores higher on all three dev sets. That tower-free configuration, not the one just described, is what F-full and the frozen MSH evaluation actually run. The first invariant still holds either way, since neither configuration has a set-identity parameter. Dropping the tower simply means the shipped model’s only route to set-relative signal is the pool tower, each candidate’s own features, and the four set-shape scalars above; it never reads the rest of the card list.

4.3 Training

One recipe for every run in the paper (fixed before the scaling and ablation studies, since the protocol forbids per-rung tuning). Batches of 8,192 picks are drawn from one (set, format) shard at a time, shards sampled proportionally to $n_{\text{picks}}^{0.5}$, an interpolation between pick-uniform (which lets the largest sets dominate) and set-uniform (which oversamples tiny sets). AdamW ($\beta_1=0.9$, $\beta_2=0.98$, weight decay 0.01), peak learning rate 10^{-3} , 2,000-step linear warmup, cosine decay; cross-entropy with label smoothing 0.05 renormalised over real pack slots. The presentation budget is four epochs. Within-training validation (never the dev sets) is evaluated every 2,000 steps and training stops after three non-improving evaluations. Seed 17 throughout. Seed variance is measured at two scaling rungs (§5).

Training runs on a single Apple M3 Ultra workstation (96 GB unified memory) in PyTorch 2.12 eager mode on the MPS backend, at $\sim 42,750$ picks/s. The full 149.4M-pick run takes about 1.8 hours, stopping at step 34,000. Training-backend correctness is not assumed and is actively gated rather than hoped for; specifics are in the companion reference. Inference exports to ONNX (opset 18) and runs on a consumer x86-64 machine, CPU-only, well under a millisecond per pick.

5 Pre-registered evaluation protocol

The evaluation is frozen at a public git tag (`eval-protocol-v1.1`) that predates the existence of the test data. The tag fixes the protocol document, the metric implementations, and the battery manifest, and a hash guard refuses to run if the metric code has since drifted from the tagged version. Model weights are too large to version directly, so each battery member’s weights are instead anchored by the sha256 recorded in that tagged manifest, cross-checked against the experiment ledger’s commit history to confirm the artifact predates the test data. There is one evaluation round; a second round would require a new public tag and disclosure that a second round happened.

5.1 Test set: MSH, exactly once

MSH is the first *Magic: The Gathering* set drafted on Arena after the protocol freeze: a licensed-IP (*Universes Beyond*) set, which makes the test deliberately harder (§7). The test snapshot is the first public 17Lands MSH Premier file successfully downloaded on or after its publication, frozen immediately by sha256 and ETag and never re-downloaded. Until the frozen evaluation has run, no training, tuning, metric-peeking, or feature-vocabulary fitting may read MSH pick data. MSH *card* features alone are legal (the zero-shot contract, §3). Quality gates at snapshot time: modern

schema, Premier rows present, at least 2,500 expert-slice drafts, $\geq 99\%$ of pack card names joining to card features. On a volume failure the single pre-declared contingency is to wait exactly one snapshot cycle. The evaluation executes within 72 hours of the gates passing: one inference pass per battery member over the frozen snapshot, cached as per-pick prediction files from which every statistic is computed. Models are never re-run during analysis.

Two populations are scored. The *expert slice* (§3) is the headline; the all-users slice is secondary. Two conditioning configurations are pre-registered: *deployment mode* (headline), conditioning on a fixed top skill bucket selected on the dev sets and frozen into the battery configuration; and *human-model mode*, conditioning on each drafter’s true bucket, scored on all users. Forced picks (final pick of a pack) are included in headline numbers, matching the per-set ceilings and prior work; non-forced variants are secondary rows.

5.2 Development discipline

One number governs every architecture, hyperparameter, text-encoder, and conditioning decision in this paper: mean top-1 agreement, on the expert slice in deployment mode, averaged unweighted over three sets (BRO, TMT, SOS), with ties broken by mean log-loss. That number comes from a single development run, *F-dev*, trained on every set except those three and MSH.

We chose that trio for three separate reasons. BRO is the same holdout used by the only published multi-set, feature-based zero-shot number [3], so our result is directly comparable to theirs. SOS is the set where our own per-set model is most mature, giving the strongest available ceiling to measure against. And TMT is a licensed-IP set, so it rehearses the same distribution shift MSH will present. Early stopping never looks at these dev metrics (§4.3), so the curve reported later in §6 is not selected on the same axis it is plotted against.

5.3 Metrics and statistics

The primary metric is top-1 agreement with the human pick, expert slice, deployment mode. Several secondary metrics accompany it: top-3 agreement; all-users top-1 (human-model mode); the *normalized score*, which divides the model’s zero-shot top-1 by the per-set ceiling’s top-1 on the same picks (a model trained directly on the target set, aligned pick-by-pick on its validation split); mean log-loss; top-label expected calibration error (15 equal-mass bins, with a temperature fit on the dev trio only, frozen into the battery before T0, and never fit on the test set itself, §7); per-(pack, pick) accuracy curves against the per-cell random floor; and *late-draft retention*, the same zero-shot-to-ceiling ratio restricted to picks 8+ of each pack, where pool context dominates.

All intervals are cluster bootstraps over drafts, not picks ($B=2,000$, fixed seed, percentile 95% CIs); paired comparisons share resample indices. Clustering matters little but is grounded empirically: measured intraclass correlation of pick correctness within drafts is $\rho = 0.0099$ (per-set SOS validation; 1,662 drafts, 69,803 picks), a design effect of ~ 1.4 at ~ 42 picks per draft, giving CI half-widths $\sim \pm 0.36\text{pp}$ at the 2,500-expert-draft gate (binomial approximation at $p=0.5$). Seed variance is measured with three seeds at two scaling rungs, and set-composition variance with two probe rungs (§6).

5.4 The battery

Frozen by artifact hash before the test snapshot: (1) **F-full**, the final recipe trained on all 31 sets, the headline model; its dev-set numbers are meaningless (it trains on them) and are never quoted. A pre-registered flexibility clause lets two optional extras (the Powered Cube file and VOW QuickDraft) join F-full’s training data if and only if a dev-trio ablation shows a gain; the decision

is frozen with the battery and reported either way. (2) **F-dev**, the development model (§above), which doubles as the top scaling rung. (3) **Scaling rungs** at 1, 2, 4, 8, 16, and 28 training sets (nested, in release order) plus two composition probes, all with the fixed recipe and a shared step cap. (4) **Ablations**: A-notext (structured features only, no text embedding) and A-noUB (the three licensed-IP training sets removed), for the shift analysis of §7. (5) **Baselines**: random; an hour-zero rarity-color heuristic; and three asterisked post-release baselines (community site ratings at day ~ 2 , and two 17Lands-statistics pickers) that bound what cheap post-release information buys. (6) **Post-day-one rows**, frozen now and executed only after the zero-shot evaluation: a per-set ceiling trained on the frozen MSH snapshot with the stock per-set recipe, and F-full fine-tuned on the MSH training split with hyperparameters pinned from dev-trio trial runs.

5.5 Claims bands

To keep interpretation out of our own hands, three mutually exclusive claims paragraphs (for headline results below 45%, between 45 and 55%, and above 55%) were drafted before any test number existed and are carried in the paper source; the matching one is selected verbatim when the frozen evaluation reports (§6). In all bands the secondary claims are the same: the normalized-vs-ceiling score, the shape of the scaling curve, the licensed-IP shift analysis, and the skill-conditioning effect.

6 Results

Zero-shot transfer on the development trio. Table 1 reports zero-shot transfer on the development trio. F-dev, the recipe used for architecture and hyperparameter selection, trains on 149.4M picks from 28 sets and is then evaluated on three sets it has never seen a pick from, where it reaches 54.3% agreement. (The paper’s actual headline result is the frozen MSH number reported below; every number in this paragraph is a dev-trio rehearsal by the protocol’s own design, §5.) Broken out per set, DraftFM reaches 48.9% / 57.5% / 56.5% on BRO / TMT / SOS. For comparison, a model trained directly on each of those sets, the ceiling, reaches 65.6% / 71.0% / 70.2%. DraftFM’s pre-registered normalized score, its accuracy as a fraction of that ceiling on identical held-out picks (§5), is 74.3% / 81.3% / 80.2% (dev-mean 78.6%). A simpler, unaligned version of the same ratio, taken over the full slice rather than the smaller aligned subset, corroborates this on a much larger sample: 74.5% / 81.0% / 80.5% (dev-mean 78.7%). That unaligned ratio is not itself the pre-registered metric, since the aligned version is capped by the size of the ceiling model’s own held-out split. Excluding forced picks lowers the three numbers to 45.2% / 54.1% / 53.2%. The deployment-mode skill bucket selected on dev is the 0.66 win-rate bucket.

Reading the BRO row. BRO is the holdout on which Bertram et al. [3] report 55.4%, the strongest published unseen-set number. They reach it with card features, learned image embeddings, and 16 usage statistics computed from human play after the set’s release, scored on an all-users, not expert-filtered, population. Our 48.9% (expert slice; 48.4% all-users, the population-matched comparator) is below that number, but the two are not a fair fight: their number assumes information that does not exist on release day. The fair comparison is against what the same paper achieves without post-release usage statistics: 33.6–35.6% on a single set, and every other day-one-feasible baseline in Table 2. Against that day-one bar, multi-set training on public card features closes most of the distance to the usage-statistics number while using only release-day information.

BRO is also the hardest set in the trio for transfer: its packs mix in a 63-card retro-artifact bonus sheet, examined directly in §7. Why TMT, the trio’s licensed-IP set, nonetheless transfers as

Table 1: Zero-shot expert-slice top-1 agreement (%), deployment mode, forced picks included; 95% cluster-bootstrap CIs over drafts in parentheses. F-full’s dev cells are never quoted (it trains on those sets). Each cell’s source run is recorded as a comment in the generated table file.

Model	BRO	TMT	SOS	Dev mean	MSH
DraftFM (F-dev, zero-shot)	48.9 (48.8, 49.0)	57.5 (57.3, 57.7)	56.5 (56.4, 56.6)	54.3	56.1 (55.9, 56.3)
DraftFM (A-noctx, selected dev recipe)	50.0 (49.9, 50.1)	57.7 (57.5, 57.9)	56.6 (56.5, 56.7)	54.8	—
DraftFM (F-full, zero-shot)	—	—	—	—	57.0 (56.8, 57.2)
Per-set ceiling	65.6	71.0	70.2 (69.8, 70.6)	68.9	TBD
F-dev / ceiling (unaligned)	0.745	0.810	0.805	0.787	TBD



Figure 3: Steel Seraph, BRO, the dev trio’s hardest transfer set (§7). Real P1P1 expected value 9.59, the 99.4th percentile of BRO’s card pool by the deployed per-set model’s own ranking; the next two candidates in the pack, Skystrike Officer and Tyrant of Kher Ridges, rank at the 99.0th and 98.7th. Percentile, not a head-to-head split: Arena hides the pack between picks in this mode (§4.2). Never MSH.

well as SOS, and why BRO lags both, is analysed there too. Figure 4 shows a real P1P1 bomb from TMT scored the same way. For sets that are in the training corpus but have no dedicated per-set model, the deployed system falls back to F-full’s scoring; Figure 5 shows what that fallback looks like today for LTR, LCI, and DMU. That figure illustrates deployment behavior, not an additional zero-shot data point.

Baselines. In Table 2, random picking scores 23.2%. An hour-zero heuristic (prefer lower rarity for playables, follow the pool’s colors) reaches 42.8%, just below GPT-4o’s 43% [2] and close to the expert-tuned 44.5% of Ward et al. [10]. That closeness is a useful caution about how much of drafting is surface regularity rather than anything deeper. DraftFM’s zero-shot numbers clear every day-one-feasible baseline in the table, on each of the three dev sets (Random and the rarity heuristic are measured on SOS as a rehearsal stand-in, not per-set). The three asterisked post-release baselines bound what cheap human data buys within days of release; their MSH rows are filled in once the frozen evaluation runs.

The frozen MSH evaluation. The headline number: F-full’s expert-slice, deployment-mode top-1 on MSH is 57.0% (confidence interval in Table 1). Snapshot gates, identifiers, and hashes are reported in the companion implementation reference (Appendix B points there).

The single cached prediction file behind that headline number also carries several secondary statistics (§5), all read from the same completed pass rather than any new run on MSH: top-3



Figure 4: April O’Neil, Hactivist, TMT, the dev trio’s licensed-IP set (§7), a comic-book bomb the text encoder never saw described in *Magic: The Gathering* rules-text style. Real P1P1 expected value 6.22, the 96.7th percentile of TMT’s card pool. Never MSH.

Table 2: Baselines and published anchors. Upper block: our measured baselines, expert-slice deployment-mode top-1 (practice rows will be replaced by frozen MSH rows by the same pipeline). Lower block: published numbers, quoted from their papers on their own (general-player, not expert-filtered) pick populations; test sets, information regimes, and pick populations all differ from the upper block and from each other, so these situate rather than compare. * = uses post-release or otherwise non-day-one information.

Method	Information used	Test set	Top-1 agreement (%)
Random	none	MSH	23.3 (23.1, 23.4)
Rarity-color heuristic	card features (hour 0)	MSH	25.6 (25.4, 25.7)
Site-ratings heuristic (day ~2)*	post-release statistics	MSH	TBD
ALSA-argmin*	post-release statistics	MSH	TBD
shrunk-GIH-argmax*	post-release statistics	MSH	TBD
DraftsimBot [10]	expert-tuned ratings	M19 (Draftsim)	44.5
GPT-4o zero-shot* [2]	LLM pretraining	NEO	43.0
Features-only, single-set [3]	card features	unseen (NEO-trained)	33.6–35.6
Features+meta+image, 13-set* [3]	post-release meta + images	BRO (unseen)	55.4

agreement is 87.7%, mean log-loss 1.132 nats, and top-label ECE 0.005 (deployment mode, expert slice). Scored on every valid pick rather than only the expert slice, agreement is 57.3%, close to the expert number. Two more secondary rows – the normalized score against a per-set MSH ceiling, and late-draft retention on MSH – need a ceiling model trained directly on MSH’s pick data, a post-day-one step we have deliberately not run yet; they stay open for a follow-up release.

Interpretation. The headline number exceeds the strongest published unseen-set result, the 55.44% of Bertram et al. [3], while using strictly less information (no post-release usage statistics, no card images) on a licensed-IP set with out-of-universe flavor (card names are masked before embedding, §4.1) – a shift the dev-trio evidence suggests text-masking mitigates but does not eliminate (§7). The BRO development row provides the same-holdout anchor against that prior. Zero-shot drafting of an unseen set is, on this evidence, better served by scale over public card features than by any published use of post-release data.

Scaling. Figure 6 and Table 3 report zero-shot accuracy as training grows from one set (NEO, 5.4M picks) to the full 28-set universe (149.4M picks), under the fixed recipe. Single-set training is



LTR: EV 6.33, pct
99.4



LCI: EV 6.10, pct
99.5



DMU: EV 5.78, pct
99.2

Figure 5: Three sets with no dedicated per-set model, Lord of the Rings, The Lost Caverns of Ixalan, and Dominaria United, all in F-full’s 31-set training corpus. This illustrates F-full’s fallback scoring path for training-corpus sets without a dedicated per-set model: the same path a newly released set would use before a per-set model exists for it. Real EVs and percentiles are computed the same way as in Figures 3 and 4. But because F-full trained on these three sets, this illustrates the production fallback path, not a zero-shot-accuracy claim; that claim is Table 1’s dev-trio and MSH rows only.

Table 3: Scaling ladder. Rung labels are frozen protocol identifiers, not set counts: *S27* names the top rung’s run, which actually trains on 28 sets, a naming artifact from an earlier corpus size, kept as-is rather than renamed after the fact. Set and pick counts in the table are the real values from run records. Dev-mean is the model-selection metric of §5.

Rung	Sets	#sets	Train picks (M)	Best step	Dev-mean top-1 (%)
S1	NEO	1	5.4	4000	48.9
S2	S1 + DSK	2	13.3	18000	50.9
S2b	MOM, TDM (probe)	2	14.5	8000	50.2
S4	S2 + DMU, FIN	4	27.8	24000	51.7
S4b	STX, SNC, OTJ, TLA (probe)	4	23.5	20000	51.0
S8	S4 + STX, MOM, BLB, TLA	8	52.9	18000	52.4
S16	S8 + AFR, SNC, ONE, LTR, WOE, MKM, OTJ, EOE	16	99.9	34000	53.3
S27	full F-dev universe	28	149.4	34000	54.3

the regime of Bertram et al. [3]’s features-only anchor; the top rung is F-dev. Two extra seeds at S1 and S4 give within-training top-1 spreads of 0.677–0.682 and 0.688–0.689. Training itself is not noise-dominated at either rung; this checks in-distribution training stability and complements the zero-shot dev-mean seed spread reported below. A log-linear fit of dev-mean top-1 on $\log_2(\text{training picks})$ across the nested ladder (S1 through F-dev, 5.4M–149.4M picks) tracks the six rungs closely: slope 1.0 percentage point per doubling, $R^2 = 0.98$, with no rung off the fitted line by more than 0.4pp. But that same fit argues against reading a diminishing-returns point into the tested range, not for one. The per-doubling gain moves between 0.7 and 1.7pp/doubling, and it does so non-monotonically: lowest across the S2–S8 middle of the ladder, highest at the S1–S2 start and the S16–F-dev end, rather than decaying toward the top rung.

Six error-bar-free points, on a fit whose own top rung is the model selected on this axis, are not strong evidence of log-linearity. The honest reading is narrower than it looks: no diminishing returns are visible over the two orders of magnitude tested, which is not the same as showing that returns do not saturate beyond that range.

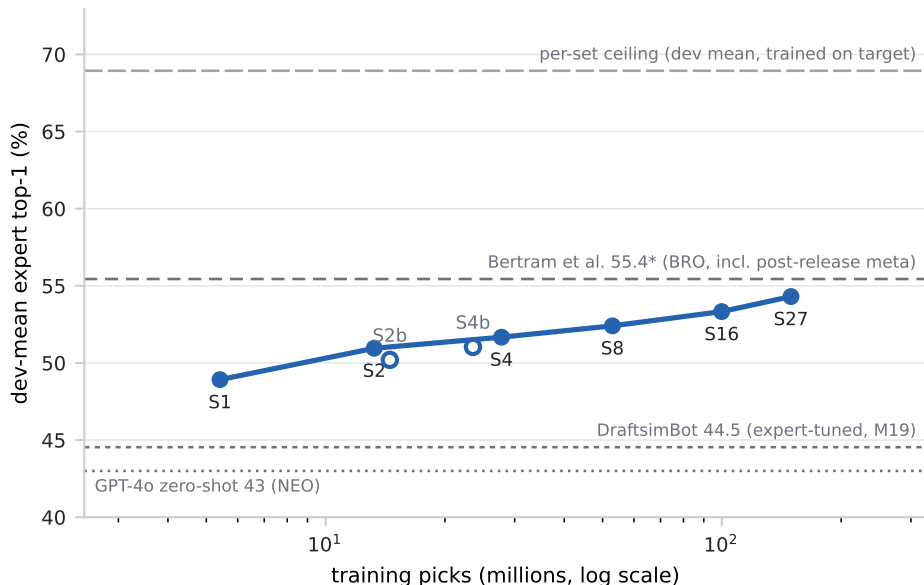


Figure 6: Zero-shot dev-trio-mean expert top-1 vs training picks (log scale). Filled points: the nested scaling ladder; open points: set-composition probes. Horizontal rules: published anchors and the per-set ceiling mean. Generated from run records by `figures/scaling_curve.py`; rungs appear as their frozen evaluations land.

Re-scoring those same extra-seed checkpoints on the zero-shot dev-mean the curve actually plots, rather than on within-training top-1, gives per-seed spreads of 0.65pp at S1 and 0.42pp at S4. That is at most about a quarter of the 2.82pp gain from S1 to S4, so the rung-to-rung climb is not a seed artifact. These are three-seed spreads at two rungs that also fold in an early-stop-step difference (the two added seeds stopped at half the reference seed’s step), so they bound the curve’s seed noise rather than fully characterize it, standing in at S1/S4 as a proxy for the top rung.

The two composition probes offer a natural check on whether pick count alone predicts a rung’s score, or whether which sets are in it matters too. At matched set count, S2b (MOM and TDM, 14.5M picks) scores 50.2% against S2’s 50.9%, despite having *more* training picks than S2’s 13.3M (Table 3). If pick count were the whole story, S2b should have scored higher, not lower. S4b (23.5M picks) scores 51.0% against S4’s 51.7%, in the direction pick count would predict, but S4 also has more picks (27.8M), so that pair does not separate the two explanations. Only S2 vs. S2b isolates set count from pick count, and on that one pair, composition beats volume: which sets a rung trains on moves the score more than how many picks they contribute. Two probe pairs is not enough to generalize beyond that.

7 Analysis

Seed variance on the top-scoring variant. Two additional seeds of A-noctx (the recipe F-full uses, §4.2) land within-training top-1 at 0.680–0.684, a 0.5pp spread around the seed reported in the table. So A-noctx’s own accuracy is not a single lucky initialization. But the margins between it and the neighboring recipe-search variants are close (54.2–54.8 dev-mean, Table 4), meaning the full ranking among variants has not itself been seed-checked.

Table 4: Zero-shot expert top-1 (%), point estimates (per-cell 95% CI half-widths are $\sim \pm 0.1$ – 0.2 pp on the dev sets, Table 1; paired-difference CIs accompany the deltas in the text). A-noctx, A-proportional, and A-topfilter are architecture variants tested against the Full baseline before the deployed architecture was fixed. A-notext and A-noUB are the two pre-registered comparisons (§5) analysed below: A-notext drops the 384-d text embedding from the Full recipe, and A-noUB removes the three licensed-IP training sets from A-noctx (the winning architecture, §4.2), so its clean comparator is the A-noctx row, not Full. The (variant $\times$ set) grid supplies the dev-side numbers for the licensed-IP text penalty (per-set deltas below), computed before MSH is seen.

Variant	BRO	TMT	SOS	Dev mean	MSH
Full (F-dev recipe)	48.9	57.5	56.5	54.3	56.1
A-notext	45.8	55.9	53.1	51.6	54.7
A-noctx (winning)	50.0	57.7	56.6	54.8	56.2
A-proportional	50.2	56.8	55.7	54.2	55.6
A-topfilter	49.8	57.3	56.5	54.5	56.2
A-noUB	49.2	57.5	54.9	53.9	55.0

The flexibility-clause extras. The protocol’s flexibility clause (§5) admits VOW QuickDraft and the Powered Cube into F-full’s training data only if a dev-trio experiment shows a gain over A-noctx. Adding VOW QuickDraft to A-noctx (same recipe and holdout, one shard added) scores 48.9/57.6/56.0, dev-mean 54.2. That is lower than A-noctx’s 50.0/57.7/56.6/54.8 on all three dev sets, and the paired-difference CI excludes zero on every set (-1.1 pp BRO, CI -1.2 to -1.1 ; -0.1 pp TMT, CI -0.26 to -0.03 ; -0.6 pp SOS, CI -0.62 to -0.52). So the clause’s own rule excludes it: F-full’s frozen recipe (§4.2) does not train on VOW QuickDraft. The Powered Cube half of the clause was not tested at all, since bulk per-pick data for it is not available through the public 17Lands pipeline this paper otherwise relies on exclusively, and acquiring it by another route was out of scope. Both extras are therefore absent from F-full’s training data, one by a pre-registered result that failed to show a gain and the other by an unmet data precondition, and the frozen recipe is exactly the 31-set corpus described in §3.

The licensed-IP shift. MSH’s names and flavor come from a comic universe, not from *Magic: The Gathering*. Its text lives in the sentence encoder’s pretraining as a different genre. This is the distribution shift most likely to break a text-dependent model on the day it matters, which is why the protocol makes a licensed-IP set (TMT) a development set: the rehearsal happens where it can still inform design. One mitigation is built in: card names are masked out of the text before embedding (§4.1), so the encoder sees mechanics, not franchises. A second design choice does not reduce the shift but makes it measurable: the comparison grid of Table 4 separates what the text embedding contributes on in-universe sets (SOS, BRO) from what it contributes on licensed sets (TMT, MSH).

Zero-shot, the licensed dev set turns out not to be the hard one. TMT reaches 57.5% against SOS’s 56.5% (and 81.0% vs 80.5% of their respective ceilings), which suggests masking plus structured features are carrying most of the load. That could mean text simply adds little on licensed sets, or it could just mean TMT is an easier format. The grid distinguishes between the two: dropping text costs TMT only 1.6pp (CI 1.5–1.7pp) against its Full comparator, less than BRO’s 3.1pp (CI 3.0–3.2pp) or SOS’s 3.4pp (CI 3.3–3.5pp). Text contributes least on the licensed-IP set, which favors the masking-plus-structured-features explanation over TMT simply being an easier format. The UB-penalty (removing the three licensed-IP training sets from A-noctx) tells a consistent story: 0.8pp on BRO (CI 0.7–0.9pp), 0.2pp on TMT (CI 0.09–0.33pp), 1.7pp on SOS (CI

1.6–1.7pp). Losing licensed-IP training data costs the licensed-IP dev set least, not most, and even TMT’s small penalty is a real, paired-CI-supported effect, not noise dressed up as a trend. If the test set is harder than the trio average, the pre-registered framing stands: a licensed-IP test makes the zero-shot claim more credible, not less.

Why BRO transfers worst. BRO sits 8.1pp below the mean of TMT and SOS, at only 74.5% of its ceiling. Of three candidate explanations, the bonus sheet looked, at first pass, like the most direct: packs offering at least one of the 63-card retro-artifact bonus sheet score 40.6% zero-shot, while packs that don’t score 55.5%, a 14.8pp raw gap on 44% of BRO’s picks.

That gap turns out to be mostly a different artifact in disguise. BRO’s booster carries exactly one bonus-sheet card, so a pack can only “have the bonus card” before it has been picked away. By construction, *has_bonus* falls from 100% at pick 1 to 2.9% at pick 15 (BRO’s last, forced pick): bonus-present packs average pick 4.1, bonus-absent packs average pick 9.3, and early picks are mechanically harder regardless of what the pack contains. Stratifying by pick number before comparing (Figure 7) collapses the gap to 3.1pp (CI 2.8–3.4), roughly 16% of BRO’s shortfall against the trio, not something comparable in size to it.

The per-(pack, pick) curve shows the same family of artifact playing out differently. The raw BRO-vs-TMT/SOS gap widens through the pack, but BRO’s booster carries one more card than TMT/SOS’s (a real extra forced pick, itself a consequence of the bonus sheet enlarging the pool), so late raw picks are not apples-to-apples across sets. Correcting for the random floor at each pick reverses the trend. Artifact-heavy color-signal weakening and 2022-era recency remain plausible explanations but are not separately isolated by this grid. With the bonus-sheet slice’s real contribution now measured at 16%, rather than assumed to explain most of the shortfall, more of BRO’s gap is left for those factors, and for anything else this grid does not cover, to account for.

Skill conditioning. Conditioning is a scorer-side input (§4.2), so one model serves any audience. In human-model mode (§5) the model scores 48.4% / 57.8% / 56.8% on BRO / TMT / SOS: within a point of the expert-slice deployment numbers, consistent with skill moving picks less than card quality does. MSH shows the same pattern: deployment mode (conditioning on a fixed top skill bucket) and human mode (conditioning on each drafter’s own bucket) differ by only 0.00pp on the expert slice, so the choice of which bucket to condition on barely matters there either.

Calibration. Zero-shot confidence falls well short of within-set calibration. Top-label ECE is 0.102 / 0.071 / 0.093 on the dev trio, against 0.014 for the within-set SOS ceiling: roughly a five-to seven-and-a-half-fold gap. The protocol permits one calibration decision: a temperature fitted on dev only and frozen into the battery, never fitted on MSH. That temperature is 1.28, chosen by minimizing mean dev-trio log-loss over a grid rather than ECE directly, since log-loss is the smooth, proper scoring rule this protocol’s other dev decisions already use. It cuts the dev-trio mean ECE from 0.089 at the unscaled $T=1$ to 0.025, closing most of the gap above. This temperature is fit on the logits of F-dev, the architecture-search model that still has the set-context tower (§4.2), and applied unchanged at test time to F-full, the tower-free recipe that actually scores MSH. F-full has no held-out set of its own to fit a temperature against without touching MSH, which is why the temperature crosses architectures this way. MSH ECE, at this same frozen temperature, is 0.005, tighter than any of the three dev-trio numbers above rather than in between them. We do not have a confirmed explanation for that on this evidence – it is not simply forced picks inflating the count of trivially well-calibrated predictions, since MSH’s non-forced ECE is close to its forced-inclusive figure – so we report it without over-interpreting it.

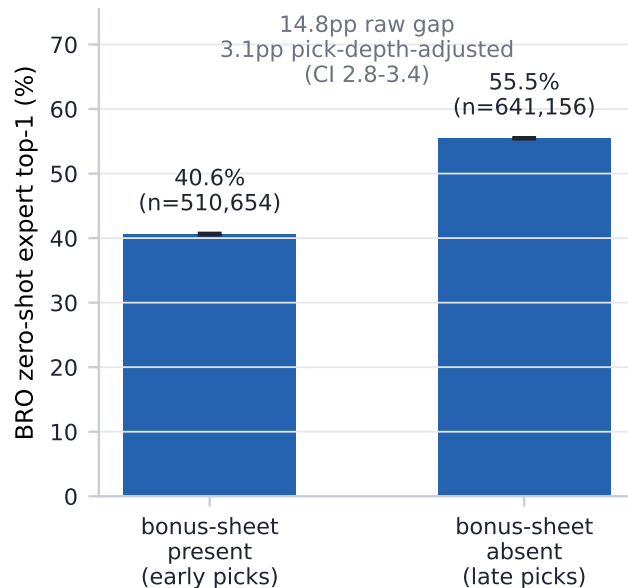


Figure 7: BRO zero-shot expert top-1, split by whether the pack offered at least one of the 63 BRR retro-artifact bonus-sheet cards (44% of BRO’s picks do). The raw 14.8pp gap is mostly a pick-depth confound (labeled above the bars): *has_bonus* tracks early-vs-late pick number by construction, and stratifying by pick number before comparing collapses the gap to 3.1pp (95% CI). Error bars on the raw bars are 95% cluster-bootstrap CIs over drafts. Generated from cached zero-shot predictions by `figures/bro_diagnostics.py`; never MSH.

Late-draft retention. Later picks are where a set-agnostic model could plausibly collapse to rate-the-card, since pool signal has taken over from the card alone. The pre-registered statistic, late-draft retention (§5), is 81.8% / 87.8% / 89.2% on BRO / TMT / SOS (dev-mean 86.3%), higher, not lower, than the corresponding all-pick ratios (74.5% / 81.0% / 80.5%, §6). That raw ratio is not, on its own, evidence that the pool tower is adding signal late. Picks 8+ carry roughly twice the forced-pick rate of the draft as a whole (forced picks score 1.0 by construction on both sides of the ratio) and a substantially smaller mean pack size (4.0–4.5 cards vs. 7.3–8.0 overall). Restricting to genuine choices, pack size ≥ 4 , excluding the trivial tail, puts late-pick top-1 *below* the all-pick figure on all three dev sets, the opposite direction from the retention ratio. The claim the data does support is narrower: the model does not collapse to chance late in the draft. The per-(pack, pick) curves against the per-cell random floor show this directly. Excluding forced picks, zero-shot top-1 never comes closer than 20.5 (BRO) / 32.5 (TMT) / 28.4 (SOS) percentage points to chance at any point in the draft, including its latest picks. [PENDING: late-draft retention on MSH]

Acknowledgments

This work would not exist without 17Lands, whose public datasets [1] are the training and evaluation corpus, and whose data-sharing policy makes reproducible drafting research possible. Card metadata is provided by Scryfall [9]. *Magic: The Gathering* is the property of Wizards of the Coast; this is unofficial research, unaffiliated with and unendorsed by Wizards of the Coast, 17Lands, or Scryfall.



Figure 8: The low-conviction end of the model’s output: a real thirteen-card SOS pack (empty pool) where Sundering Archaic and Flow State score a near-exact tie (EV 1.41 vs. 1.41, dominance split 50/50) – the third candidate, Pursue the Past, trails at 27% dominance against the pair. Ties this close are exactly the regime the top-label ECE below is measuring: the model’s stated confidence and its actual reliability at low conviction. Computed by the deployed per-set model, never MSH.

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A Limitations

The target is agreement, not optimality. Top-1 agreement with expert humans measures imitation. Experts disagree with each other (the within-set ceiling is $\sim 70\%$, not 100%), and agreeing with them is not the same as maximising win rate. A model could in principle draft better than its agreement suggests, or worse. Win-rate evaluation of zero-shot drafters is future work.

Population. 17Lands users are self-selected: they run a tracker, they draft on Arena, and the expert slice further conditions on volume and success. All numbers are relative to this population; paper drafting, other platforms, and casual play are out of scope.

One evaluation round. Pre-registration buys credibility at the cost of adaptivity: the frozen battery cannot react to surprises in the test snapshot. And a single set is a sample of one from the distribution of future sets. The dev-trio spread and the composition probes bound, but do not eliminate, how much the headline number could vary on the *next* set.

Text and language. The text encoder is English-only and frozen. Rules text is embedded as prose, so mechanics whose meaning is mostly numeric or positional may be under-represented. Name masking removes the most direct pretraining leak for licensed sets, but flavor vocabulary in the remaining text still differs by universe (§7).

Numerics. Training runs on an accelerator backend with documented silent-error classes; we gate on CPU parity and skip non-finite steps (§4.3), which guards the classes we know about.

B Reproducibility

Every result in this paper traces to a run id in an append-only ledger. Every table cell and prose number is emitted by a script from that ledger and cached evaluation outputs; no result number here is typed by hand. The ledger schema, frozen identifiers (protocol tag, manifest hashes), release contents, and script-level pointers for reproducing any specific number are in the companion implementation reference, not repeated here.

C Licensing and attribution

Draft data. Draft data is from the 17Lands public datasets [1], used under CC BY 4.0 and per the project’s usage guidelines (bulk S3 downloads; no API scraping).

Card metadata. Card metadata is from Scryfall [9], used per its API guidelines; Scryfall neither produces nor endorses this work.

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